

Project Proposal: Sleep, Stress & Lifestyle Insights

A Data-Driven Biomedical Research Project for the Irvine Community

Problem & Community Relevance

Poor sleep and high stress are widespread concerns among students and families, contributing to reduced academic performance, weakened immune function, and long-term health risks. While Irvine-specific health data is not publicly available, national and international sleep research consistently shows the same risk factors likely affect local students and families. This project analyzes public biomedical datasets to identify which lifestyle and physiological factors most strongly relate to poor sleep and elevated stress, then translates those findings into a practical, shareable tool and recommendations for the Irvine community.

Research Question

How are sleep duration, sleep quality, and lifestyle factors (physical activity, caffeine, alcohol, exercise, BMI) associated with stress level, sleep disorders, and sleep efficiency?

Hypothesis: Individuals with shorter or lower-quality sleep will show higher stress levels, higher rates of sleep disorders, and lower sleep efficiency than those with longer, higher- quality sleep.

Data Sources

Two public, de-identified Kaggle datasets (an unrelated animal-sleep dataset was excluded): 1. **Sleep Health & Lifestyle Dataset** (374 records) — sleep duration/quality, stress level, physical activity, BMI, blood pressure, heart rate, daily steps, sleep disorder diagnosis. 2. **Sleep Efficiency Dataset** (452 records) — sleep duration/efficiency, sleep-stage percentages, awakenings, caffeine/alcohol use, smoking status, exercise frequency.

Methods

1. **Data cleaning** — remove duplicates, fix data-parsing errors, handle missing values.
2. **Exploratory analysis** — visualize relationships (e.g., stress by sleep-duration group).
3. **Predictive modeling** — for each target (stress, sleep quality, sleep disorder, sleep efficiency, awakenings), train and compare a naive baseline, an interpretable linear/logistic regression, and a random forest, evaluated on held-out test data.
4. **Interpretation** — use feature-importance rankings to identify the strongest predictors.

Deliverables

- A public, interactive website (Streamlit, already deployed) for exploring the data, comparing model accuracy, and testing personal lifestyle inputs against the models.
- A full written report documenting methodology, results, and limitations.
- A one-page recommendation sheet for Irvine students and families on sleep and stress.

Ethics & Limitations

- Data is public and not collected in Irvine — findings represent general patterns, not local statistics. All results describe **associations**, not proof of cause and effect.
- The project and predictor tool are educational only, not a diagnostic or medical tool.

Timeline

Week	Task
1	Finalize datasets, clean and prepare data
2	Exploratory analysis and visualization
3	Build and evaluate baseline models
4	Build interactive web app; deploy online
5	Write final report and community recommendation sheet; present findings